

1. The 3×3 reduced row echelon matrix derives from the original matrix A 's pivots. Given that this is a square matrix, the R matrix contains only columns such that every other element is zero. The d vector contains the solution to $Ax = b$.

* Question 1.c. was asking for numerical values. Points will not be subtracted as they are found here (circled)

-5 $Ax = b$
 $x = A^{-1}b$

$\Rightarrow d = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 12 \\ 20 \end{bmatrix}$

$b = \begin{bmatrix} 4 \\ 12 \\ 20 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix}$

-9 $\Rightarrow \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -4 \end{bmatrix} \Rightarrow$

-8
b. the column space consists of two independent columns such that their combinations fill the rest of the columns.

As such,

$C(A) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 2 \end{bmatrix} \right\}$

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ can be formed from the column space above.

3. A is a 3×5 matrix

a. Then the equation $Ax = b$ has many solutions.

Using the nullspace solutions:
 $c_3x + c_4x + 0 = 0$
 $2c_3 + c_4 = 0$
 $5c_3 + c_2 = 0$

* there is only one pivot.

$\Rightarrow A = \begin{bmatrix} 1 & -2 & -5 \\ 3 & -6 & -15 \\ 5 & -10 & -25 \end{bmatrix}$

$\Rightarrow R = \begin{bmatrix} 1 & -2 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

E_{31}, E_{21}

Since only column 1 contains a "free" variable, the only special solution is:
 $\begin{bmatrix} c \\ 0 \\ 0 \\ 0 \end{bmatrix}$ where c is any real number.

As such, the nullspace are all combinations of the first column plus zero times each other column. (counts only for $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ 2,2,2)

-5 $B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$

$\Rightarrow \begin{bmatrix} A & A \\ 0 & 0 \end{bmatrix} \Rightarrow$

$\begin{bmatrix} 0 & 1 & 2 & 2 & 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 & 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 & 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Asking for free! Double! A what?

4. $A = \begin{bmatrix} 2 & 5 \\ 6 & 4 \\ 2 & 4 \end{bmatrix}$ Since we know the solutions to $Ax = b$,

these form the column space by which any combination (addition) give the remaining solutions. This is because they form their own space by which solutions can be constructed.

b. the column space of A consists of three independent columns that form the remaining columns. the nullspace consists of the zero vector plus the amount of special solutions - which would be 2 given the number of free variables.

b. Yes!
 $3a_1 + 2a_2$
 $\Rightarrow 3 \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 26 \\ 14 \end{bmatrix}$

with $r=3$

always

worked differently but makes...